

‘A Fair Game?’: Using Narrative Sensification and Embodied Metaphors for Awareness

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Figure 1: Left: A render of *A Fair Game*. Right: A player at the installation.

ABSTRACT

Most physical data representations rely on vision only and miss the opportunity of making people physically experience data. This work introduces ‘A Fair Game’: a narrative sensification that uses story-telling and embodied interaction to provide data insight. A Fair Game is a two-player horse race installation representing passport power, where people pull on a rope to move their horse. People randomly draw a high- or low-ranking passport which determines the horse’s speed, so either a lot or little physical effort is needed to win the race. The installation was deployed at an exhibition for two days, where we conducted interviews with fifteen people. Findings indicate that people’s embodied interaction and the visible

unfairness of the game contributed to emotional engagement. Participant’s felt frustration and empathy, which facilitated space for discussions regarding the topic. Our work shows the potential of kinesthetic and emotional engagement, and introduces the concept of narrative sensification.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; Visualization techniques.

KEYWORDS

data physicalization, narrative physicalization, embodied interaction, critical design, decolonial perspectives

*These authors contributed equally to this research and would like to disclose their nationalities. The nationalities are (in order of the names): Colombian, Pakistani, Lebanese / Ukrainian, and Indian.



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1 INTRODUCTION

To this day, one of the most persisting narratives fostering global inequality is the concept of ‘developing’ and ‘developed’ countries [51]. Historically, such narratives are shaped and controlled by what is considered the ‘center’: Global North countries which often used to be colonial powers, and institutions belonging to or stemming from that center [3, 11, 17, 51]. As a result, the story of ‘developing’ and ‘developed’ countries has been framed by a Western-centric, post-colonial lens, which neglects the diverse cultural, social, and historical contexts [11]. As pointed out by Loomba, the post-colonial implies a temporal aftermath, suggesting that colonialism has been overcome or ended [3]. This way of thinking overlooks the fact that colonial inequities and power dynamics continue to shape the world to this day.

A concrete example of such inequity is passport power [28, 88], where certain passports gain access to more countries than other passports. For example, according to the Henley Passport Index (HPI) 2023, a German passport provides visa-free access to 190 countries, whereas a passport from Bangladesh only gives visa-free access to 40 countries. Thus, people born in the ‘right’ places will have an easier time travelling and moving to other countries, have less costs and delays resulting from having to apply for visas, and can stay abroad longer [70]. All these aspects can be a major advantage when it comes to studying and working for international companies [88], and have even resulted in passport investment migration, where citizenships can be bought by making large investments [46].

However, problems such as passport power are difficult to grasp and understand, as: (1) depending on their nationality, people have different experiences with the topic (which can make it difficult to imagine what it is like for others), (2) people do not like to be reminded of their privilege and acknowledging that others do not have it, and (3) passport power is a so-called *wicked problem* where there is not one easy solution, as it stems from a complex, non-linear web of multiple causes and actors [13]. This makes it difficult to grasp and intangible. Passport power is an example of a topic where activists and organisations, as well as educational institutions such as museums, use data to aim to educate the public about issues of societal importance (such as climate change, gender and racial discrimination, and other concerns of social justice). Yet for lay people, numerical data often are abstract [15] and it is easy to ignore if the topic is uncomfortable.

An approach for conveying data that has been gaining traction in the research fields of HCI and Data Visualization (DataVis) is *data physicalization*: physical data representations that encode the data through their geometric or material properties [41]. Physicalizations have been used in various contexts (e.g., art and museum installations, design work, academia) to convey data and make them concrete [6, 36]. Physicalisations have been particularly popular to illustrate data about complex issues of societal relevance [64]. For example, the wicked problem of climate change is a common topic for physicalizations (e.g. [22, 72]), as they offer emotional experiences that go beyond the qualities of rationality, efficiency, and accuracy offered by standard visualizations [63, 83, 84]. This can be experienced in the physicalization “*Slave Voyages*” [49], a set of two artifacts that represent the harrowing subject of the Atlantic

slave trade. The aesthetic and situated qualities of this physicalization helped in creating awareness and enhancing (emotional) engagement with the data on atlantic slave trade.

However, physicalizations can represent data beyond the aesthetics of the visual modality. In their recent overview of design variables for data physicalisation, Hornecker et al. [33] highlighted the potential for encoding data in properties experienced through the kinesthetic senses, and that rely on bodily interaction. In these cases, visual inspection of the artefact or object does not reveal the data (even if there are subtle variations in shape). Recent research on physicalizations that demand active interaction with the artifact has shown that engaging people’s kinesthetic sense helps to enhance and foster social engagement with the data [38]. There are two types of physicalizations which engage people’s kinesthetic sense: data sensifications [29] and narrative physicalizations [43]. Similar to physicalizations, narrative physicalizations use physical artifacts to represent data. However, whereas physicalization do not necessarily tell a story with the data or use narrative elements (e.g., 3D bar graphs [78]), narrative physicalizations make the user experience the data along a story arc, which helps to situate the topic and provide context which helps people understand abstract data [43].

However, even though the use of narrative arcs and kinesthetic interaction in physical data representation appears promising, little research –outside that of Hurtienne et al. [38] and Karyda [43]– has explored their effects on our perception and understanding of data. This coincides with another research gap in data physicalization. Although research has begun to investigate how physicalizations impact user experience beyond traditional measures of DataVis, such as efficiency and usability (e.g., [57, 71, 84]), there is relatively little on the impact of physicalizations which go beyond the visual modality. This probably is partially due to the fact that most physicalizations focus on the visual modality [6, 33, 63]. When research does explore this, it tends to focus on multi-sensory data representations (which use two or more sensory modalities to represent data) [32], which are not necessarily data physicalizations [16, 31]. For example, Lee et al. introduced the concept of *data visceralization* in Virtual Reality to evoke “*visceral feelings*” regarding a data set (e.g. the distance between the Earth and the Moon) to make its understanding more intuitive as opposed to the traditional numeric perception of data [53].

Therefore, to explore the effect of kinesthetic interaction in a physical data representation, our work combines the ideas of narrative physicalization with data sensification in the interactive installation called *A Fair Game*: a two-player narrative sensification of passport power data. The aim of *A Fair Game* was to explore how data interpretation through sensification can contribute to a better understanding of complex issues, while identifying valuable elements that enhance the user’s data experience. To this end, *A Fair Game* was deployed during a public exhibition at the Bauhaus-Universität-Weimar, during which people from inside and outside the university experienced the sensification. Of all visitors who interacted with the installation, fifteen people took part in interviews regarding their experience with the sensification. From our observations and interviews we learned that *A Fair Game* made the complex and intangible topic of passport inequality experienceable

and graspable for both people with higher and lower-scoring passports in real life. The combination of embodied experience and the visible unfairness of the sensification created feelings of validation, empathy, and discomfort with the unfairness. This then opened the space for sharing experiences and questioning the unfairness of real-life passport inequality.

With this work, we contribute to a better understanding of physicalizations that go beyond the visual modality, by introducing the concept of narrative sensifications. Through our case study with *A Fair Game*, we illustrate the power that experiencing the meaning of data through one’s own embodied actions can have and how this can provide a more visceral, direct understanding of the implications for those affected. The multi-modal nature of the sensification (kinesthetic and visual) and narrative arc fostered emotional engagement, which helped to make people critically reflect on the data and create space for discussions. Our findings show the potential of narrative sensifications for representing wicked problems and could help to push for more physicalizations that go beyond the visual, by including embodied interaction.

2 RELATED WORK

2.1 Data Sensification

Data sensification is a class of data representation that aims to generate insight “by encoding data in the behavior, functionality, performance, or affordance of an object” [29]. Similar to the concept of ‘data visceralization’ (where one can bodily experience the data, e.g., in VR) [53], active interaction with the representation is required to experience the data encoding and obtain data insight. An example of a data sensification is “*The World’s Best Spintop*”¹ [61]. Melanie Bossert created a number of 3D-printed spinning tops, whose shape and behavior are a representation of the respective country’s political, environmental, health, education, quality of life, and economic data. Based on data input for these topics, the shape of the spinning top is generated by the algorithm. Countries that perform ‘poorly’, receive an irregular shape and a short handle, whereas countries that perform ‘well’ have a symmetrical shape and a handle that is long enough to grasp. Based on these form factors, well-performing countries have spinning tops that are easily put in motion, remain in balance, and spin for a longer time than countries with a lower performance score, of which the short handle and unsymmetrical shape make it difficult to put and keep the spinning top in motion. Although the shape of the spinning top already gives an idea of how the country scores on the underlying data, subtle difference in symmetry are difficult to identify from the object alone, and it is only by putting the spinning top in motion that we are able to experience and understand the full data.

Where other data representations rely on a specific representational modality (e.g., visualizations rely on the visual and sonification on the auditory modality), in sensifications the data are encoded through metaphors, as opposed to a specific modality [32]. “*The World’s Best Spintop*” uses the metaphors of stable motion (through balance) to represent how a country performs.

Sensifications can be seen as a sub-field of *data physicalization*: physical artifacts that represent data through their geometric or

material properties [41], and multi-sensory data representations [32] which combine two or more representational modalities to represent and communicate the data (e.g., sound and vision). Key to sensifications is that they rely on data being conveyed through a multi-modal interaction with a physical artifact [29, 38], which is where they differ from physicalizations, which can convey data without multi-modal interaction (e.g., relying on vision alone). Because of the required active interaction, sensifications can create immersive experiences and greater awareness about the topic at hand [38, 43], as the user engages in an unfolding narrative which turns them into an active participant of the story [69]. Moreover, sensifications rely on embodied interaction, building on a deeply entrained kinetic-sensorial understanding of the world [33, 60], such as their experience of balancing oneself and balancing objects. This way, sensifications “enhance people’s reflective process” [43] and bridge the gap “between data and emotion” [69].

Because of these characteristics, previous research has highlighted its potential for creating awareness for wicked problems [69] and *data activism*, which aims to generate “positive social, institutional, environmental, and economic change” [76]—thus, making sensification a possible suited method for representing passport power.

2.2 Narrative Visualization and Physicalization

Another category of data representation that requires active interaction are *narrative physicalizations*, a type of physicalization that borrows from narrative visualization [43, 73]. Similar to data physicalization, narrative visualizations have been used for centuries (cf. Nightingale’s Rose Diagrams) and only relatively recently, has the use of storytelling gained traction in DataVis [18, 73]. Narrative visualization employs narrative techniques to tell a story with data, where the visualization guides the viewer in unpacking and understanding the story. Commonly used techniques are visuals that situate the data (the aesthetics of the visualization fit the data), event sequencing (structured time-course of events), and having a story arc with a clear beginning, middle, and end [35, 55]. Research has shown that narrative visualizations help viewers to better understand the data, engage audiences, foster emotional connections, and provide the necessary context to the data [23, 34, 44]. The use of narratives to present data has even been successfully used on its own, without visualization, in the “*Data Epics*” project, where sensor data were represented as stories [14].

Inspired by the advantages of data narratives, narrative physicalizations use similar techniques, but instead of 2D visualizations, they use physical artifacts or interactive installations to create tangible representations of the data narrative [43]. Furthermore, through active interaction, narrative physicalizations situate the data in a particular experience which allows the user to bodily recall or undergo the lived data experience. For example, Karyda et al. created three narrative physicalizations based on people’s hobbies, which represented their heartbeat or step count data. To experience this data and get data insight, participants had to play Foosball or a custom-made instrument along a visualization of their heartbeat data, or walk on a treadmill to re-experience their step count data [43]. This way, narrative physicalizations engage users or audience members in an embodied, experiential, and multi-sensory manner,

¹Pictures and more information about the project can be found here.

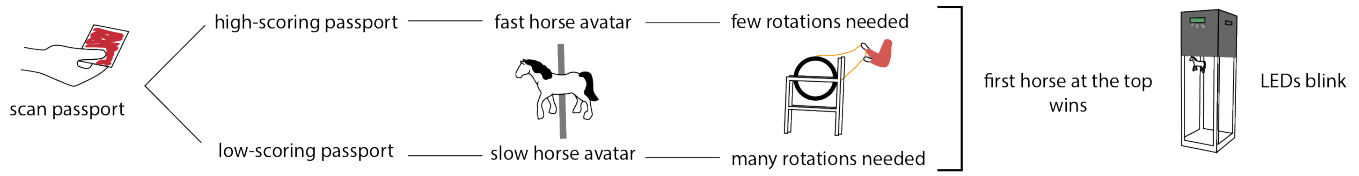


Figure 2: Overview of the process behind *A Fair Game*. Based on the passport, the stepper motor either makes the horse avatar move fast or slow. The speed of the horse thus influences the amount of rotations needed to win the game. Rotations are measured with a hall sensor and a magnet attached to the wheels. When a horse reaches the top, LED lights blink.

which can lead to a deeper understanding and emotional connection with the data [43, 43]. Compared to static data representations, this has the added benefit that the user experiences enhanced reflection, as they are part of an ongoing synthesis of information [43]. Because of this enhanced reflection, our work explores the impact of a *narrative sensification* on how people experience, make sense, and understand the wicked problem of passport inequality.

2.3 Decolonial Computing and Showroom Research

Awareness has increased for the fact that despite the formal dismantling of colonial structures, the echoes of colonialism endure within cultural expressions, practices, and knowledge systems [3, 4, 11]. Looking for post- and/or decolonial literature, also in the ACM digital library and its interactions magazine (cf. [85]), reveals a growing body of work. In Human-Computer Interaction (HCI) research has –amongst others– discussed computing’s “colonial impulse” [17], envisioned decolonial technologies and praxis (e.g., [66, 74, 79, 86]), and created a manifesto for decolonizing HCI research [5].

It is important to address the distinction between post-colonial and decolonial. As explained by Ali [3], the term *post-colonial* indicates the end of colonialism [40], whereas *decolonial* acknowledges that we are still dealing and unpacking the results and effects of colonialism [68]. It is at this point that we position our research. Decolonial thinking can expose the operation of power and examine the *who*, *where*, and *what* that contribute to the upholding of power structures [17, 51, 65]. As passport power is an example of one of the many colonial remnants, we used a decolonial perspective to create a narrative sensification to make people experience and aware of this issue, for which we drew upon *showroom research*² [48].

Showroom research borrows from the arts and critical design [7, 21], where design is used to make people reflect on the futures they want and is deployed in exhibition settings—rather than university labs or field settings [48]. As design provides a “script” to people of what the user and the product are supposed to do [2], the critical design uses “*Verfremdung*” (or defamiliarization/strangeness [7]) as well as tensions to provoke reflection and debate [21, 48]. In our research, we used the strangeness of a rigged game to provoke people to reflect and think about passport inequalities.

²Although we borrow from showroom, this article uses terminology more common in HCI and field research (such as *data* instead of “*returns*”) to keep in line with common HCI vocabulary and embrace the study’s mix between showroom and field research—as commonly done [48].

3 ‘A FAIR GAME?’—THE INSTALLATION

The work discussed in this article centers around the installation *A Fair Game* (Figure 2). *A Fair Game* is a two-player game inspired by horse races and coin-operated horse racing arcade game machines³, and is meant to demonstrate the power behind passports. Similar to arcade horse racing games, the players of *A Fair Game* compete to reach the goal first, here by pulling their horse up (Figure 3), using a bicycle wheel as their controller. Players stand around the ‘track’, a vertical, rectangular construction, each at their own input console, where they pull on a rope. Pulling the rope has the corresponding horse figurine move up the track. Each rope is held by and wrapped around a suspended bicycle wheel, which rotates with the rope (to create an infinite loop). The rotation of the wheel is then transmitted to a motor that pulls the player’s horse up.

However, contrary to most ordinary games, *A Fair Game* is rigged. To start the game, each player first has to randomly draw a passport (actually, a folded piece of paper) from a bucket. Each passport corresponds to a country listed in the Henley Passport Index 2023⁴. This index ranks the passport power of 199 countries, based on the number of other countries a passport gives access to. For example, Singapore holds first place, as it gives access to 192 other countries. On the other hand, a passport from Somalia (ranking 99th) only provides access to 35 other countries. Players need to first scan their passport, thus ‘checking in’ with their passport ranking. This then determines the racehorse’s speed: a high ranking passport in the passport index has the horse move faster than a low ranking. A player with a higher ranking passport thus needs to rotate the wheel less often than a player with a lower ranking passport, who needs to invest a lot more effort for the same resulting distance—making it far more difficult to win the game. Thus, the speed and effort needed to move the racehorse metaphorically represent the power imbalances, privileges, and unfair nature of born citizenship and passports. Our data mapping uses what Hornecker et al. call *visual and haptic explicit variables* [33]. We utilize the explicit visual variables of *speed* (the speed of the horse moving upwards encodes the player’s passport power) and the haptic variable of *kinesthetic force* (the player feels the bodily exhaustion needed from the extra effort/force to move the horse). When exhibiting the installation, we ensured there was always one player with a lower-scoring passport (from the last ten countries on the HPI) and one with a high-scoring passport (one of the top ten countries of the HPI). Therefore, drawing the passports was not fully random, but made sure the game was rigged.

³Example can be found here and here.

⁴<https://www.henleyglobal.com/passport-index/ranking>

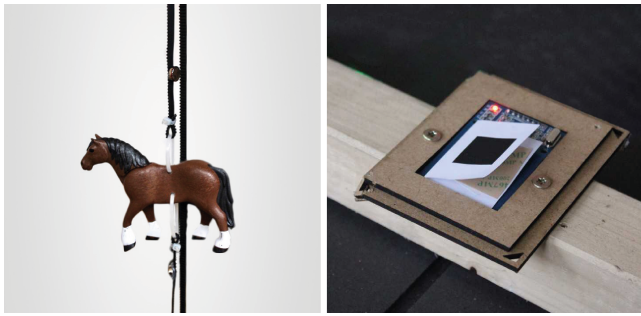


Figure 3: In *A Fair Game*, players moved their toy horse (depicted left) upwards by pulling a rope (thereby rotating a wheel). The player whose horse first reached the top won. The right image shows the passport check-in, where the paper passport is placed on a RFID reader.

3.1 Technical Details

A Fair Game was designed using easy-to-get electronic parts and controlled using Arduino. Stepper motors were used to move the toy horse avatars up and down. To register if and when a horse had reached the top, a small magnet was attached to the pulley belt (see Figure 3—left). A hall sensor was then used to indirectly measure the distance between the horse and the end of the race track. If the magnet was close enough—meaning that the horse had reached the top—the LED lights at the top of the installation lit up to show the winner (Figure 4), and the direction of the stepper motor’s rotation was reversed to send the horse down again.

For the passport check-in system, each paper passport contained an NFC tag (Figure 3—right). To start the game, participants had to ‘scan’ their passport, by holding it against the RFID reader, attached to the bottom of the race track, next to their horse avatar. The check-in then determined the speed of the horse. Low-scoring passports had a speed of 200 rpm and high-scoring passports 4000 rpm.

To measure the number of rotations participants made using the controller, each wheel had a magnet and a hall sensor. The hall sensor then measured the rotations, where one full rotation was mapped to one step from the stepper motor. This, together with the horse’s speed (either 200 or 4000 rpm), resulted in a major effort imbalance between high and low ranking passport holders. The code and schematics for the installation can be found in the supplemental materials

4 STUDY APPROACH

A Fair Game was deployed at the annual winter exhibition ‘Winterwerkschau’ of the Bauhaus-Universität-Weimar in Germany (Figure 5). This exhibition offers students and employees of the university the chance to display their works, allowing visitors from within and outside the university to experience the topics of research and teaching and students’ work. As the winter exhibition draws in a variety of visitors—academic and non-academic, young and old, different nationalities, etc.—*A Fair Game* was deployed here, as this gave us access to a broad audience that could interact with the installation and could volunteer to be interviewed about their experience. *A Fair Game* stood in a room that hosted two other

installations at the exhibition for two days, during which 25 people interacted with the installation. This corresponds to roughly 75% percent of visitors who entered the room. Figure 4 visualizes our approach.

At least two researchers were present at all times with the installation to: (1) ensure that the installation was working properly so people could play the game and assist visitors in this, and (2) to take observational notes. Visitors who showed interest in the installation were asked by one of the researchers whether they would like to play the game. In pairs, visitors were then told to each draw a random passport, which either corresponded to a passport with a high or low score on the Henley Passport Index. Without knowing the meaning behind the passport, the ‘players’ scanned their passports on the RFID Reader of their input console, after which the game started. As described in Section 3, the player with the high-scoring passport has an easy race (only needing a couple of pulls/wheel rotations to send the horse to the top of the race track), whereas the player with the low scoring passport needs to invest a lot more effort to achieve the same result. Once the players were done with the game and one of them had won, they were debriefed by one of the researchers, who explained the purpose of the installation: showing the unfairness of passport power in the real world. Players could then (if they wanted) leave their thoughts and feedback on a whiteboard using sticky notes.

Furthermore, the players were asked whether they were interested in taking part in an interview about their experience with the game. When the players agreed, a consent form with information regarding the interview data gathering and processing was given to players to read and sign. Moreover, the process of interview and data processing was explained verbally to them. Once visitors consented, the semi-structured interview began. During interviews, participants were asked: (1) which country/passport they had drawn; (2) how they experienced the game; (3) how they thought the passport affected the race; (4) their thoughts on there being ‘weak’ and ‘strong’ passports; (5) whether and how interacting with the installation changed their feelings towards this topic; and (6) whether the installation was effective in getting the message across.

In total, 15 people consented to be interviewed. The interviewees were between the ages of 20 to 30 years old and hailed from Global South and Global North countries, with a total of ten countries. There were 3 participants from Germany, 3 from Nigeria, 2 from Vietnam, and one each from Trinidad and Tobago, India, Libya, Spain, Sudan, Peru and China. Interviews lasted on average 20 minutes (min. 10, max. 45 minutes). Based on participants’ preferences, the group constellation, and general situation, the interview was either conducted with an individual, with a pair, or with a small group. Seven interviews were conducted individually, two interviews were done with pairs, and one with a group of three participants. The participants of the pair and group interviews were either friends or in a relationship. The pair/group interview ensured that these participants felt comfortable and did not have to wait for each other (as they would probably want to go on and see other exhibits).

Finally, observations of the player’s interactions with the installation, the other player, and their reactions to the game (body language and comments) were written down whilst visitors to the exhibition were interacting with *A Fair Game*. The players ranged

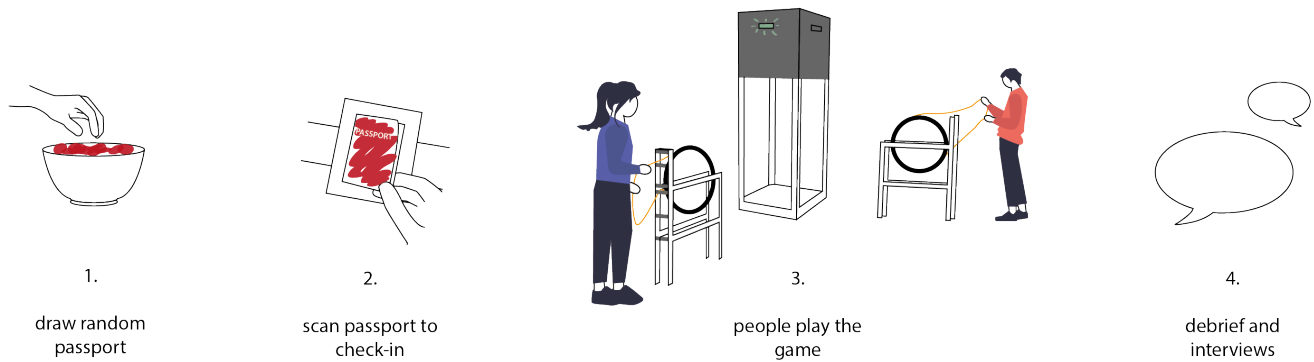


Figure 4: People first drew a random passport (1) which they used to check in and start the game (2). The passport either belonged to a lower or higher ranking country. Once there was a winner, the LEDs on top of the installation light up, indicating the winner (3). Afterwards, people were debriefed and asked whether they wanted to take part in an interview (4).

from children (accompanied by parents) to middle-aged adults from various countries. As it was crucial to observe people’s natural interactions with the installation, consent for making observations could not be obtained beforehand. This is a common for field studies in non-controlled public settings (cf. [12]). Observations were noted down completely anonymous, where we only referred to, e.g., ‘person 1 interacting with the game’ and only took notes of grown-ups interacting with *A Fair Game*.

4.1 Data Analysis

The interview transcriptions, observational notes, and feedback from visitors on the whiteboard were analyzed using Reflexive Thematic Analysis (RTA)—a process in which areas of text are assigned codes to generate common patterns among the multiple sources of data, as well as to note exceptions and other interesting phenomena [8], while acknowledging the researcher’s role in shaping the interpretation. For the analysis, we followed Braun and Clarke’s six-phase framework [8]. The analysis was conducted by the first three authors of this article, who started by each individually familiarizing themselves with the data to get an initial understanding of its content. Subsequently, over the course of one week, each researcher individually generated initial codes by identifying and labeling information that was relevant for our understanding of participants’ experience with *A Fair Game* and their understanding of the data/topic. Once the initial codes were created, the three researchers came together and collectively discussed their initial coding, to talk over any differences. When the researchers felt they had an unanimous understanding of the initial coding, they together organized, structured, and grouped the codes into five themes and seven sub-themes, each containing three to eight codes. The organizing, structuring, and grouping of codes was a process of a week, after which the initial thematic map was discussed with the other authors of this article. After a reflection period of three months, the identified themes were refined by the first five authors of this article, resulting in four distinct and unique themes and six sub-themes. The themes were then renamed accordingly and presented to the last author of this article, who had been uninvolved in the refining iteration.

5 FINDINGS

One of the aims with the deployment of *A Fair Game* was to explore whether a narrative sensification can be used to create awareness about passport power and to investigate which elements affect the user’s understanding. Throughout the deployment, people appreciated the playful aspect of the installation, as it helped to lighten the complex topic of inequality: “It’s a dark topic, but it made it a bit more light.” (I2) and “I really enjoyed how playful it was, because it really helped reach a topic which is so controversial and serious in a rather lighter way” (I5). Moreover, as the sensification offered an embodied experience, participants appreciated the low cognitive load needed to understand the topic: “Instead of it being a lecture for three hours or some posters and pie charts, it was something that people immediately realized” (I5).

Besides an understanding of participants’ overall experience, four themes were constructed in the thematic analysis. These themes show how participants’ experience and understanding unfolded through their interactions with *A Fair Game*.

Initially expecting the installation to be fair and similar to a ‘normal’ race game, the difference between assumed performance and outcomes moved participants from expecting an innocent game towards an awareness of the topic and purpose of the game. The physical exhaustion that people with low-scoring passports experienced had an important role in creating awareness for the systemic inequality behind passports, as they had to put in way more effort than their opponent. This physical exhaustion often resulted in emotional frustration, which –according to participants with a low-scoring passport in real life– mimicked how they sometimes felt when having to travel. The visible unfairness of the game combined with the physical exhaustion and emotional frustration created awareness for the topic, as people who are unaffected in real life could either see or experience the frustration. This resulted in broader discussions where participants shared their own experiences and addressed the consequences of passport inequality.

In the next sub-sections we describe how participants moved from expecting an ‘innocent’ game, experiencing physical exhaustion as well as frustration when their efforts were not rewarded to questioning the passport system, as well as experiencing feelings of



Figure 5: *A Fair Game* in use during the exhibition, where both children and adults interacted with the installation.

validation (if they held a low-ranking passport themselves) or empathy with those against whom the game was rigged. Used quotes are indicated with an *O* or *I* prefix to indicate whether they stem from our observations or from the interview respectively, followed by a participant number (e.g., I4). Quotes taken from the anonymous feedback sticky-notes are marked by SN.

5.1 From Innocence to Awareness

The exhibition context of *A Fair Game* reminded people of festival installations: “When I first saw it, I felt like it’s something you see in a festival” (I3). This situatedness [87] combined with the game-like nature of *A Fair Game*, created the illusion of an innocent race: “I saw that mine was going quicker than participant x, I felt [...] good to be honest because I was like: yeah, I’m gonna win” (I3).

However, once people started playing the game, they soon noted differences in the effort needed: “Why does her horse not move at all?” (I9). Participant’s initial reaction to the dissimilarity in speed and performance was that the installation might be malfunctioning: “I thought like maybe it’s not working or something so that’s why I had to stop to make sure that it’s working” (I12). However, once they realized they had checked in with different passports, participants started to suspect that the game was rigged: “I thought it was based on my ability [...], but when I saw that it didn’t matter, my personal strength or my personal will, [...] like the factors was (sic.) just beyond me, I was kind of disappointed.” (I3)

This led to participants wanting to change or ‘upgrade’ their passport in order to increase their chances of winning: “I want to change my citizenship for success.” (O6). One participant with a weaker passport jokingly implied that they will marry the player with the stronger passport: “I will marry you” (O9), and a father of two children who were playing asked whether there were equal passports for both of his children: “Is there a better passport for my daughter?” (O8). In one instance, the participant even demanded help from the player with the stronger passport: “You should help me!” (O14).

The visible and experiential unfairness of the game forced participants –also those who are unaffected in real life– to think about the inequalities of passports: “I never really thought about this. I don’t know if it’s because [...] I’m German. I don’t know. But I never thought about passports having different types of power” (I3), and helped them realize the role of privilege: “Seeing that visually as a race puts into context how all of life is a race and we were all trying to get to the same place. And now I get to see that we’re actually [in] the same race, different speeds, different reasons, different backgrounds.” (I3).

5.2 Physical Exhaustion and Emotional Frustration

Key to creating awareness was the difference in people’s embodied experience. Players with a low-scoring passport needed to pull the rope to turn the wheel more times, compared to players with a high-scoring passport who only needed around a couple of wheel rotations before their horse reached the top. As players with a high-scoring passport needed less effort to move their horse upwards, these players expressed how easy it was for them to secure victory: “I didn’t struggle” (I12). The ease in performance quickly made them realize that they had a chance at winning: “when you see the first time your horse goes up and how fast it goes, then you realize that you have more opportunities”(I6), which further motivated them: “you are just more excited about it because you feel that you can do it” (I6).

In contrast, players with weaker passports voiced their frustration: “I played really hard from the start, tried really hard, but it still cannot change the result compared to the other player” (I8) and “frustrating to see things working for someone else and questioning why it’s not working for me” (I7). This made them feel stressed and anxious: “I feel stressed, anxious a bit and frustrated. Like keep trying, keep trying, but it doesn’t move” (I8).

The frustration and visible unfairness made participants feel uncomfortable, as it gave the impression that not everyone is equal: *“I’m a strong believer of equality. So I feel like, why should this passport have more privileges than the others? I don’t really feel good about it.”* (I10)

5.3 Validation and Empathy

The unfairness of the game resonated with those participants that hold weaker passports in real life, who felt validated as the game revealed their personal experience that their efforts do not garner the same results as those of people with a better-scoring passport, even if they put in the same or more effort: *“I felt validated. I felt vindicated that more people are being aware because there is a difference between just feeling that you are being treated unfairly and the complete denial of that there is any unfair treatment going on.”* (I5). They liked that people with stronger passports are made aware of the inequality and can even experience what it feels like to have a weak(er) passport—even if it is only in a game setting: *“It validated my feelings. Like, showed me that: yes, this is what’s happening. This is what I’m going through. I was glad that now maybe he [the opponent with stronger passport in real life] knows”* (I12).

Some participants who had strong passports in the real-world expressed empathy for people who did not have this privilege and that *A Fair Game* made them recognize and think about their privileges: *“they have to go through things harder than someone else who just casually get[s] to go through life, I think is really eye-opening.”* (I3). This notion was further reinforced by a participant recognizing their own privilege as someone with German nationality: *“I’ve never thought of traveling being something inaccessible.”* (I3).

This empathy also translated into the game. Players holding the stronger passport often felt compassion towards their opponent: *“I felt compassion for my wife because she’s my wife and not doing well”* (O1). One player with a strong passport even paused their effort for a moment to give their struggling opponent a chance to catch up: *“I stopped because the other player was struggling”* (I12).

5.4 Questioning the System

Finally, *A Fair Game* resulted in participants questioning and criticizing the system in various ways. Upon experiencing the physical exhaustion and emotional frustration, participants who held lower-ranking passports in real life expressed how difficult it can be, to be affected by conditions that are not yours to change: *“as a citizen or as a human being, I shouldn’t be affected by the policies and politics and economy of my country. [...] I’m just a citizens (sic.), I don’t have any hand in that, so it feels unfair to some extent and... it’s just sad”* (I12) and *“[it’s] the country’s fault, not mine”* (I9).

This questioning led to the exploration of power dynamics beyond the unfairness behind passport rankings. For example, how well a country interacts and is connected with other countries in the Global North: *“if it’s [your country] within good terms with like controlling countries like the USA or like Germany and so on, you’re gonna be in a good position. If not, or you’re just an African country [...], you’re gonna be in a bad situation”* (I12), and pointing out that these are remnants of colonialism: *“what’s funny in this, that countries at the top of this table are the reason for this (sic.) country’s sufferings”* (I3).

As *A Fair Game* helped people who are normally not negatively impacted by their passport and nationality to realize that there are hardships for those who are affected, the installation contributed to creating a space in which people could share their experiences. These experiences dealt with various aspects of systemic inequality, such as language requirements: *“I [Vietnamese] have to spend the whole year of learning the language [German] and then another year of just taking a preparation course. But if I was Korean or Japanese, I don’t have to”* (I8), unfair housing opportunities: *“People that are [...] not from European countries don’t get fair opportunities in finding apartments, because they prefer Europeans or people with strong passports.”* (I12), medical restrictions: *“I have a friend who cannot donate blood in Germany because she’s from a country where she’s more likely to have malaria. It’s practical but it’s a reminder.”* (SN), and the classism in passport inequalities: *“If you are coming from a low-income background [then] you cannot afford to follow these visa procedures and go through all these”* (I3).

Participants even reflected on how passport inequalities foster power misuse: *“It is so common in tech companies in India to use the Schengen visa as a carrot dangling in front of their overworked employees or the US passport or the Canada passport.”* (I5), and how these force people to look for opportunities elsewhere: *“some people have to actually migrate for their children so that their unborn children could have the powerful passports and so that they don’t have to suffer the same way that their parents did.”* (I12)

6 DISCUSSION

With our case study of *A Fair Game*, the aim of this article is to show and discuss the potential of narrative sensifications for creating awareness and facilitating discussion of wicked problems. Our findings align with that of previous work in data sensification, which found that sensifications foster discussion and social engagement [38]. As discussed in Section 5.4, through the interaction and experience with *A Fair Game*, people’s own embodied experiences and emotional connections help them to relate the complex (and, for some, distant) topic of passport power to themselves, and created a space where people could discuss the topic of passport power and the bigger system of power dynamics.

In our discussion, we specifically touch upon the role of emotions and embodied experience in our narrative sensification, followed by a reflection on narratives and metaphors.

6.1 The Emotional Value of Narrative Sensification

Traditionally, the field of Data Visualization (DataVis) concerns itself with representing data accurately and efficiently [50, 80, 81]. Through commonly used visualization principles –such as the lack of ornaments, geometric shapes, and 2D views– [45], data are represented in visualizations which should be easy to read for visualization literate people. However, the functioning of and positivist mindset behind traditional data visualization has been critiqued—both from within and outside the DataVis community (e.g., [1, 34, 44, 45, 69]). For example, the visualization principles contribute to the *“god trick”* [26]: the feeling of being able to see everything from nowhere, and post-hoc rationalization, where people initially feel something or connect to visualizations (as shown

by [30]), but post-hoc rationalize their experiences in order to ‘objectively analyse’ the data [82]. The latter makes it difficult to allow and acknowledge the role of emotions in data visualization, even though research has stressed their importance in viewing, understanding, and analysing visualization [30, 44, 82].

Therefore, new ways of representing data can be useful to open up data experience and provide space for emotional aspects. For example, research into data physicalization discusses how physicalizations enhance our connection to the data [41], foster emotional engagement [84], and challenge people to broaden their understandings of data [6, 36, 63, 83]. As physicalizations do not rely on the same strengths and principles as visualizations [83], people do not feel that they should rationalize their experiences—as is the case for visualization, where people have learned and been taught to analyze these representations objectively [82]. Instead, the newness of physicalizations makes that they can be explored freely, triggering reflection on what the data mean and allow emotional connections to the data [58, 63, 84].

To foster engagement with the data, our work borrowed from narrative physicalization and visualization, where it is common to first “set up or base reality” [23]. In other words, sets the stage for the story to unfold. For *A Fair Game* we did so by relying on people’s experience with (arcade) games and horse races. As this is something most people either have experience with or know from popular culture (e.g., movie references), people entered *A Fair Game* expecting to be playing a common race game against each other. Having set the stage, the next phase of the narrative data representation is “struggle, complication, or development, pointing towards a climax” [23]. In our case, the unfairness or rigged nature of the game literally had participants struggle to move their horse up towards the finish line. This elicited both feelings of frustration and empathy. Once participants realised the game was not malfunctioning but rigged, they enter the ending or climax of the story. In our case, we left this deliberately open, to give participants space to reflect and discuss.

By adding a story arc to our sensification, our work introduces the concept of *narrative sensification*, where the user becomes a protagonist in an unfolding story through their interactions with the data representation [29, 43]. The unfolding story (and embodied experience, as we will reflect upon soon) elicited emotional experiences in the participants: they expressed frustration regarding their situation in the game and empathy for the other player, and generally for those who are affected by passport power, both in the game and in real life. Their behaviours whilst playing the game reflected this frustration, with examples of participants suggesting to marry or talk their way into a better passport. Therefore, it seems that the emotional connection elicited by *A Fair Game* was crucial in helping participants grasp the concept of passport power and its consequences.

As our work is an initial study showing the role of emotional connection in narrative sensification, we see a lot of directions for future research. On the one hand, we need a better understanding of the differences in emotional connection for various data representations. For example, how does people’s emotional connection differ for data physicalizations and narrative visualizations? Or do narrative sensifications result in higher or lower emotional engagement than data physicalizations? Moreover, we need to understand how

emotional connection impacts our understanding of the data. On the other hand, it would be fruitful to gain a better understanding of emotional engagement in narrative sensification: what are the element which foster emotional connection? How can we design for these elements? And what levels of emotional connection are desirable for enhancing our understanding of the data and topic at hand? These are only some of the questions that could be worthwhile to explore. Therefore, future research should look into this, to get a better understanding of the value and role of emotional connection in data representation.

6.2 Embodied Experiences in Data Representation

Besides emotional connection, participants’ embodied experiences were crucial in developing an understanding of passport power. As discussed in our findings, the physical effort participants had to exert worked as a metaphorical mapping to represent the extra effort needed to deal with the consequences of having a lower ranking passport. Thus, in *A Fair Game* participants could not only feel the data (as is the case with data physicalization) [58], but also get a glimpse of what it feels like to be *affected* by the data. Here lies the strength of data sensifications: whereas physicalizations can represent data without actually requiring direct touch, narrative physicalizations [43] and sensifications [29] need to be interacted with, in order for the user to obtain any data insight. Based on our findings, this embodied experience appears to be valuable for better data understanding.

The value of embodied experiences in learning has been extensively researched within psychology, neuroscience and has been applied in the context of HCI as well [9, 24, 25, 59, 89]. Referred to as *embodied learning*, this field recognizes bodily interactions as a way to enhance cognition [19] and is based on the idea that we come to understand the world around us through repeated embodied experiences [62]. Embodied learning is developed throughout our lives, and works for both long-term as well as salient short-term experiences [47]. The latter could have been the case for participants’ interactions with *A Fair Game*.

Embodied learning has been used to help people understand complex and abstract topics, such as physics [75] and mathematical equivalence [39], or in the domain of data visceralization, to elucidate scales or relative speed given the subjective illusion of being present in virtual environment [53]. In case of *A Fair Game*, participants first-hand, active embodied experience seem to have contributed to understanding passport power. Therefore, it seems to be valuable for future data representations to make use of embodied learning and explore the role of the body in data representations—something which is currently underexplored [16, 31]. Here we see a major potential for more research into data sensification—as to our knowledge, only two scientific articles have yet explored this (cf. [29, 38])—as well as data physicalizations which go beyond the visual modality (another current research gap, cf. [6, 16, 71]).

6.3 The Role of Narratives and Metaphors in Sensification

In *A Fair Game*, the narrative of the installation invited people to become the protagonist of the data story. The moment participants

‘chose’ their passport, they entered the narrative of the sensification, which they needed to interact with in order to experience data insight. For this narrative, we relied on metaphorical mappings—something common for sensification [29]. For example, we use the metaphor of “*life is a race*” as the first anchoring element of our horse race game, and the horse’s speed / effort needed to move the horse is a metaphor for the passport score (its power).

Research into metaphors explains how they help us make sense of the world around us [52], as they allow us to map unfamiliar topics to familiar ones. In our case, passport power (an unfamiliar topic to a large body of people who are not affected by it) was mapped to a rigged race game, where the unexpected behavior of the game triggered reflection on the topic. Because of metaphors’ power to map the unfamiliar to the familiar, they are extensively used in HCI, design, and visualization research [37, 54, 56, 67]. Metaphors also have an essential role in sensifications [32] and data physicalization [20], and they seem to have been crucial in our work to help people understand passport power. Therefore, we deem it useful to further explore the roles metaphors play in physical data representations. For example, which topics or types of topics benefit from metaphors in physical data representations, are there certain modalities which benefit more from metaphorical mappings, do metaphors improve our understanding in physical data representation, and what are their limitations?

Besides metaphors, the overall narrative of our sensification played an important role in participant’s engagement and understanding of the topic. Narratives are part of the *imaginaries* we have [42, 77]: the narratives we tell ourselves, the metaphors we use, and the mental models we rely on [56]. These imaginaries influence how we approach issues that affect us. For example, when our imaginaries are that climate change is a serious issue, we are likely to perform more sustainable behaviour, such as recycling our trash [56]. As narrative data representations influence and help us build certain narratives [34], we see their potential for guiding people in their understanding of complex issues (such as passport power and other decolonial topics) and impacting how we respond to these issues. At the same time, this persuasive power means that there are serious ethical considerations when creating and using narrative data representations—something which is known from visualization research [10, 27, 34]. Therefore, future research should explore the potential and role of narratives in data representation, but at the same time look into its ethical side to prevent misuse.

6.4 Limitations

Our study was conducted over the duration of two days, located in an exhibition space at the university. Because of this situatedness and duration, our work has some limitations. Firstly, as participants only shortly interacted with *A Fair Game*, we do not know the long-term effects of our interventions. We only know that participants expressed and showed awareness whilst being at the exhibition, but it is unknown whether this awareness persists for a longer period of time. Therefore, future studies are needed which include baseline testing and the long-term effect of interacting with a narrative physicalisation. Secondly, despite deploying *A Fair Game* at an open exhibition, most of the participants who came to visit and interacted with the game had ties to the university, as they were

studying or working there, or befriended with people from the university. Therefore, more research with a more diverse and larger sample is needed to explore the generalisability of our work.

7 CONCLUSION

In this work, we introduce a narrative sensification which demonstrates the effects of passport ranking through a decolonial lens. We live in a world where remnants of colonialism exist to this day, however, these affects are often difficult to grasp and make understandable to those unaffected by them. An example of this is passport power, where certain passports provide (easy) access to more countries than others. To represent passport rankings and the power they hold, we created a narrative sensification called *A Fair Game*, which consisted of a two player horse race inspired by coin-operated arcade racing games. People controlled a toy horse by rotating a wheel—having to put in physical effort to move the horse. To start the game, people had to draw a random passport, which corresponded to either a low or high scoring passport. This determined the speed of the horse, where high scoring passports would have faster moving horses and thus required less physical effort to win the game.

A Fair Game was deployed at an university exhibition for two days, during which we conducted interviews with fifteen people regarding their experience with the installation. Our findings indicate that *A Fair Game* helped to create awareness for the topic of passport power. Not expecting a rigged game, participants moved from ‘innocence’ to awareness. The embodied interaction and exerted effort made people feel what it is like to be (un)affected by passport power. This leveled the ground for discussion, as people got an idea of what it is like experiencing someone else’s reality. Furthermore, the imbalance between needed effort and the visible unfairness of *A Fair Game* elicited frustration, empathy, and feelings of validation in participants; contributing to emotional engagement with the topic.

Our work shows the potential of using embodied interaction and kinesthetics in providing data insight, and the value of emotional connection—things which are currently under-explored in data representations. Furthermore, we introduce and show the potential of narrative sensifications, whose multi-modal and storytelling qualities helped to represent data regarding the wicked problem of passport power. We contribute to a better understanding of the power that experiencing the meaning of data through one’s own embodied actions can have and how this can provide a more visceral, direct understanding of the implications for those affected.

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